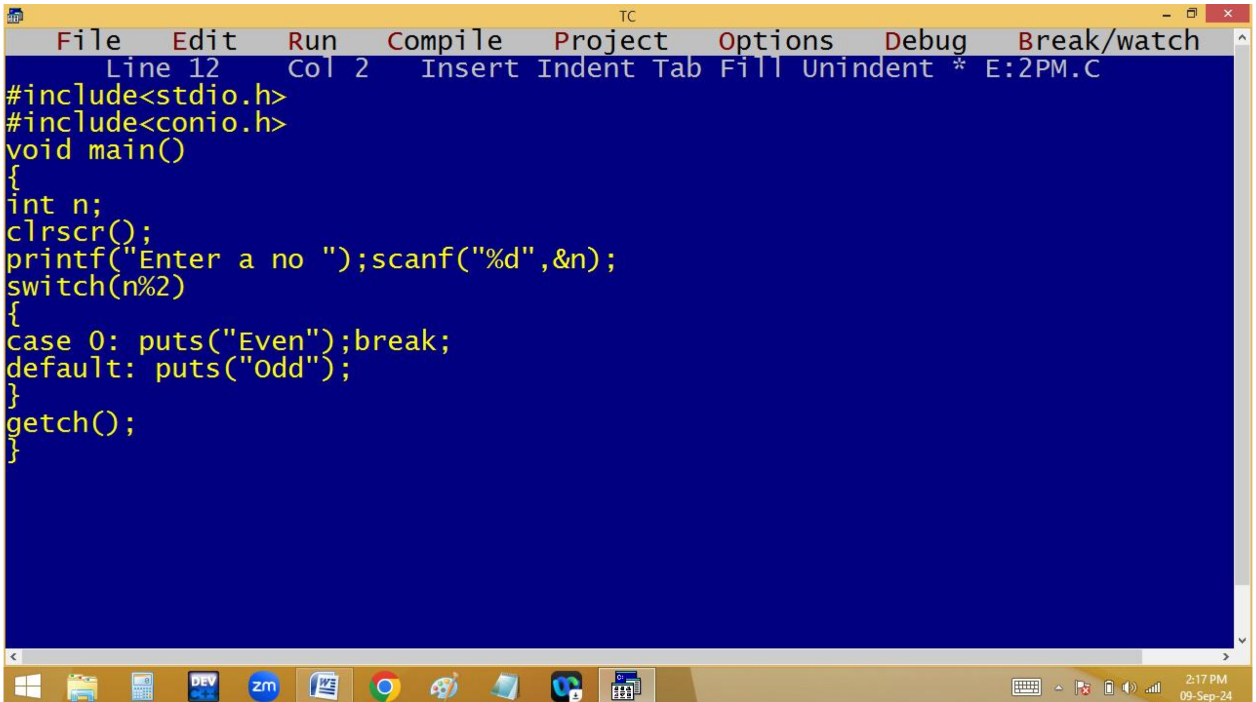
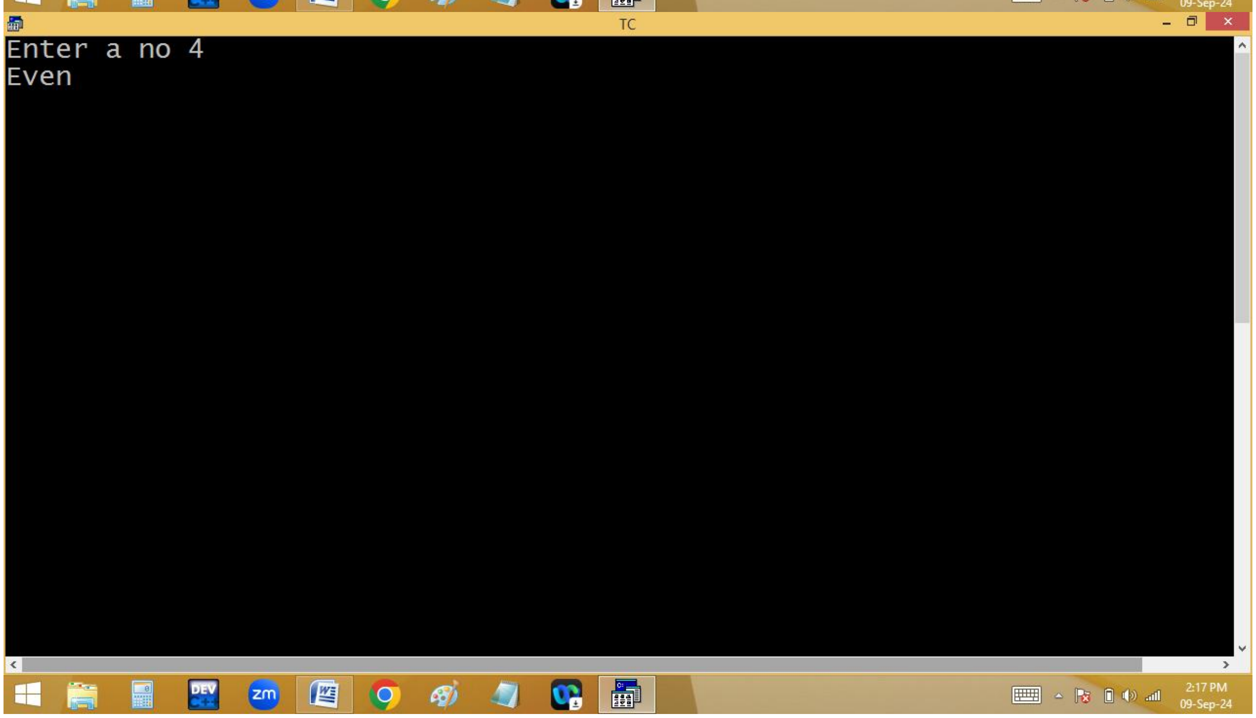
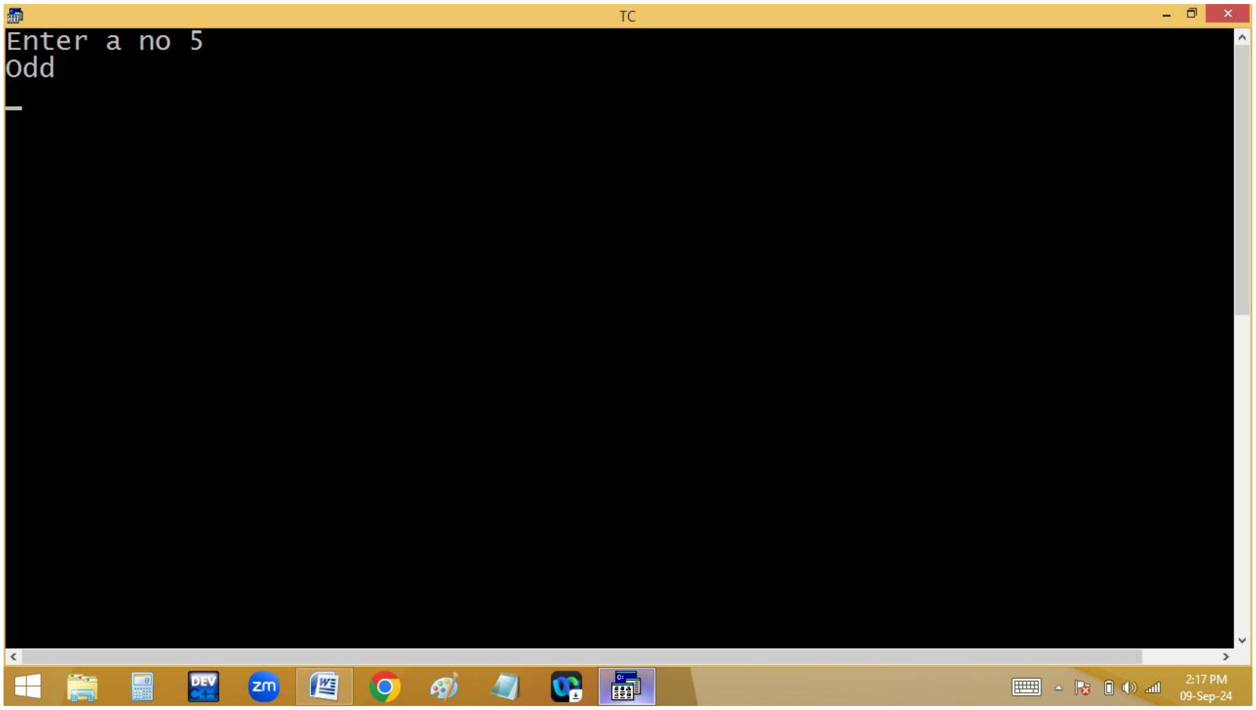
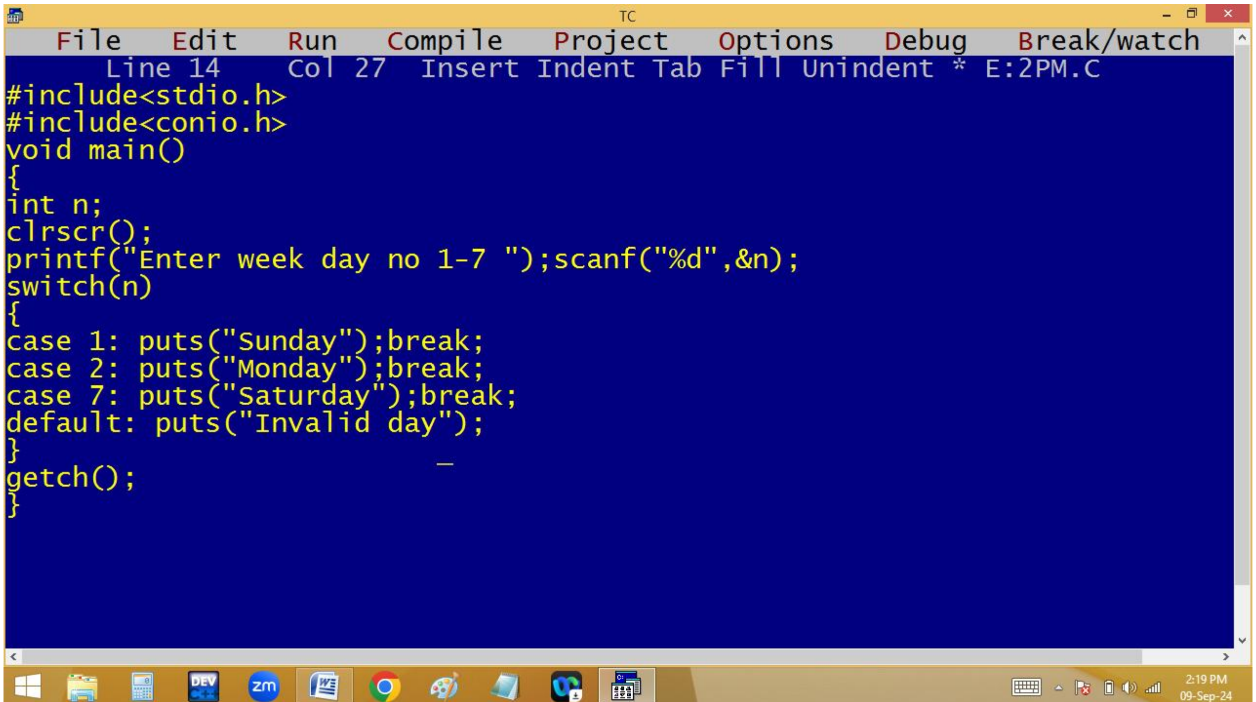
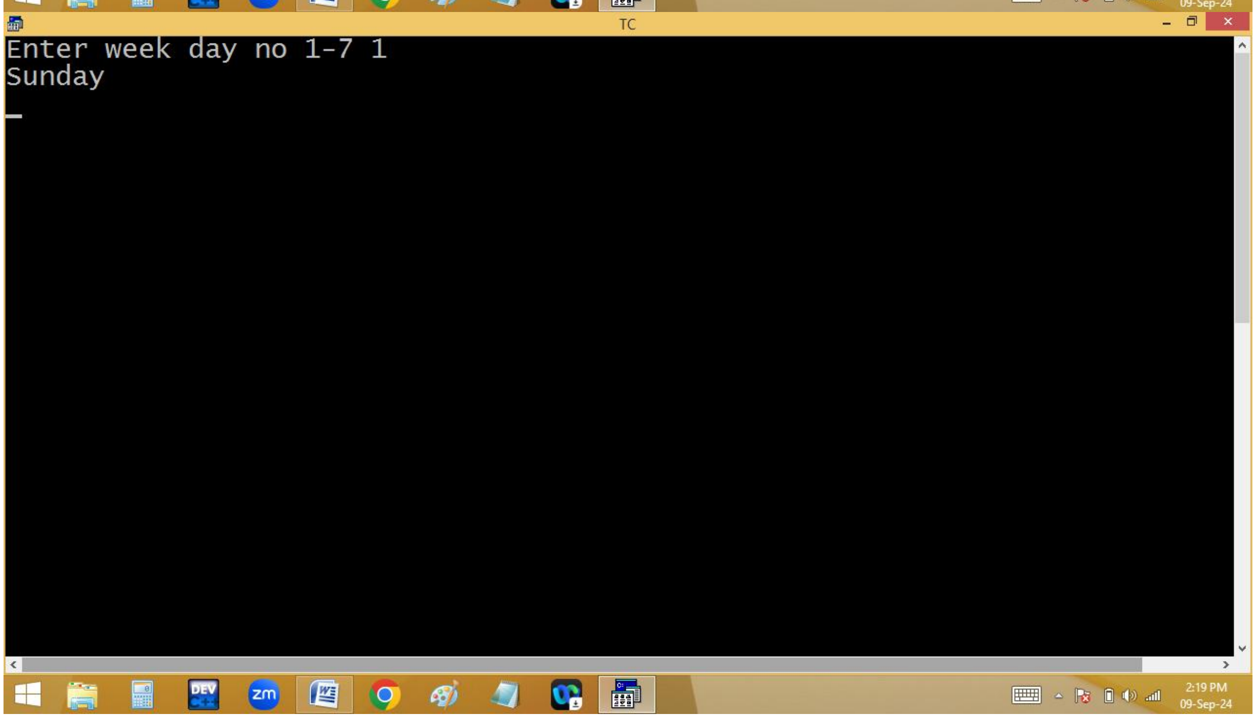
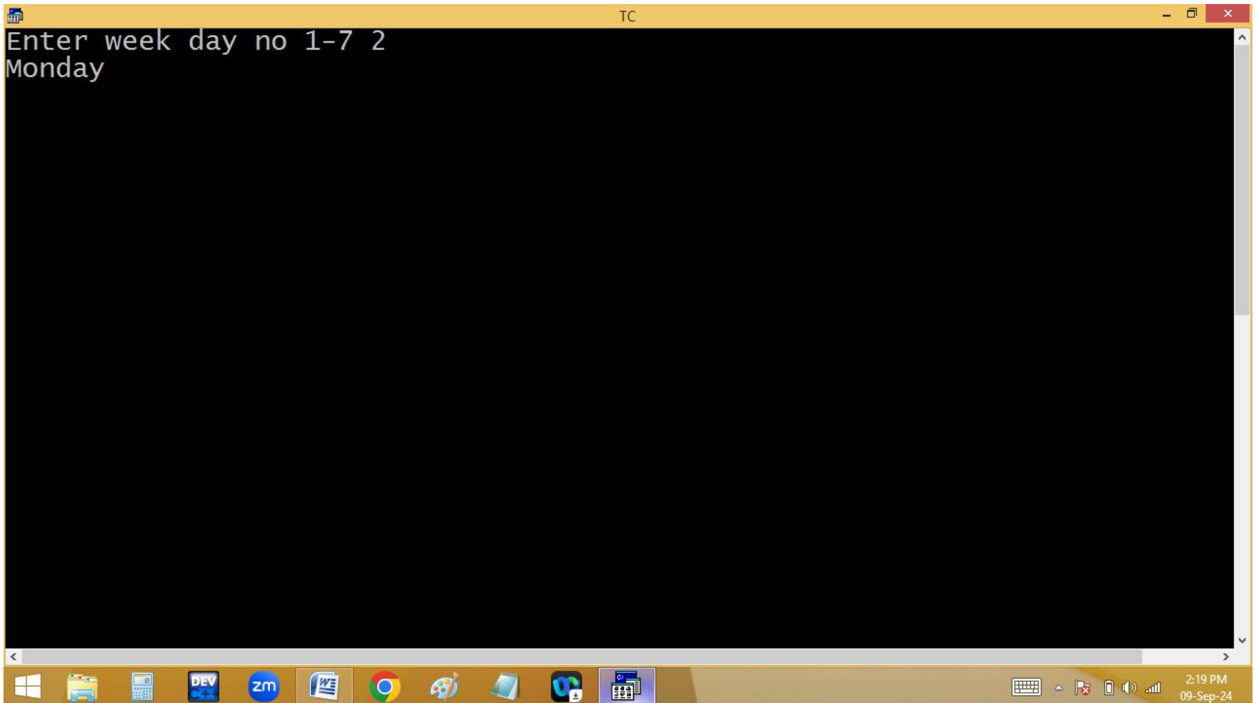
Finding even/odd using switch:

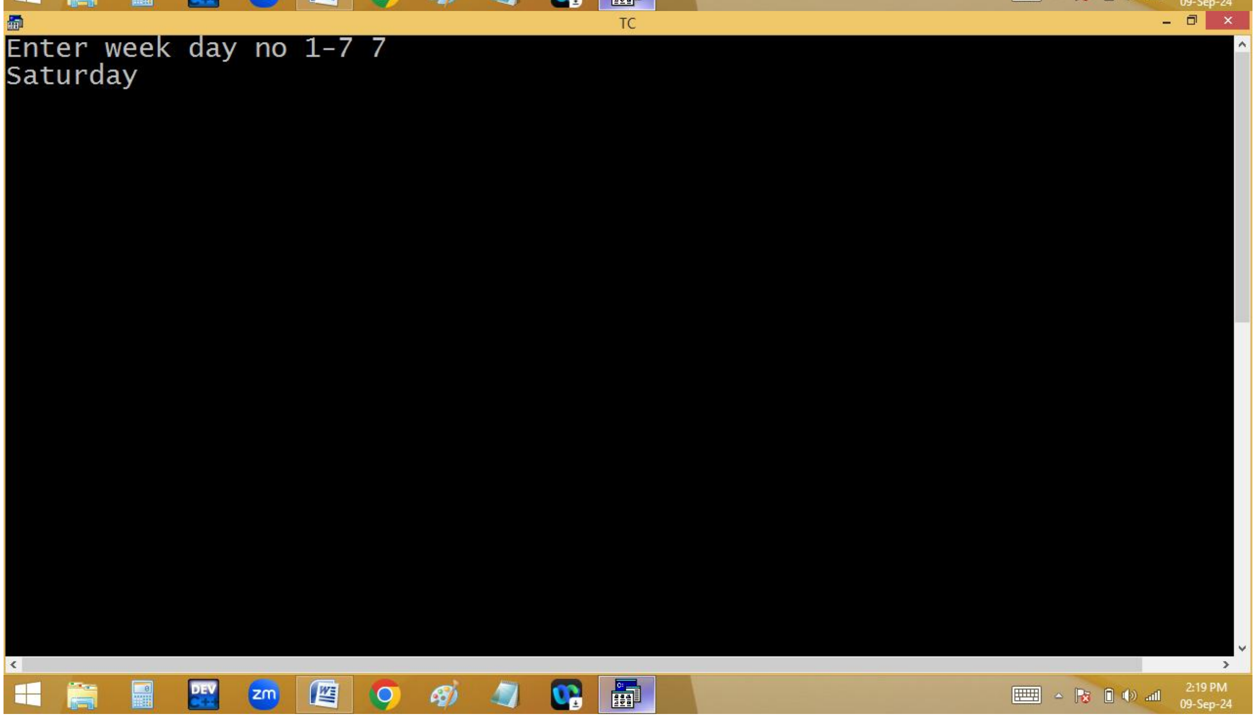

TC
File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 2 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter a no ");scanf("%d",&n);
switch(n%2)
{
case 0: puts("Even");break;
default: puts("Odd");
}
getch();
}
2:17 PM
09-Sep-24

TC
Enter a no 4
Even
2:17 PM
09-Sep-24

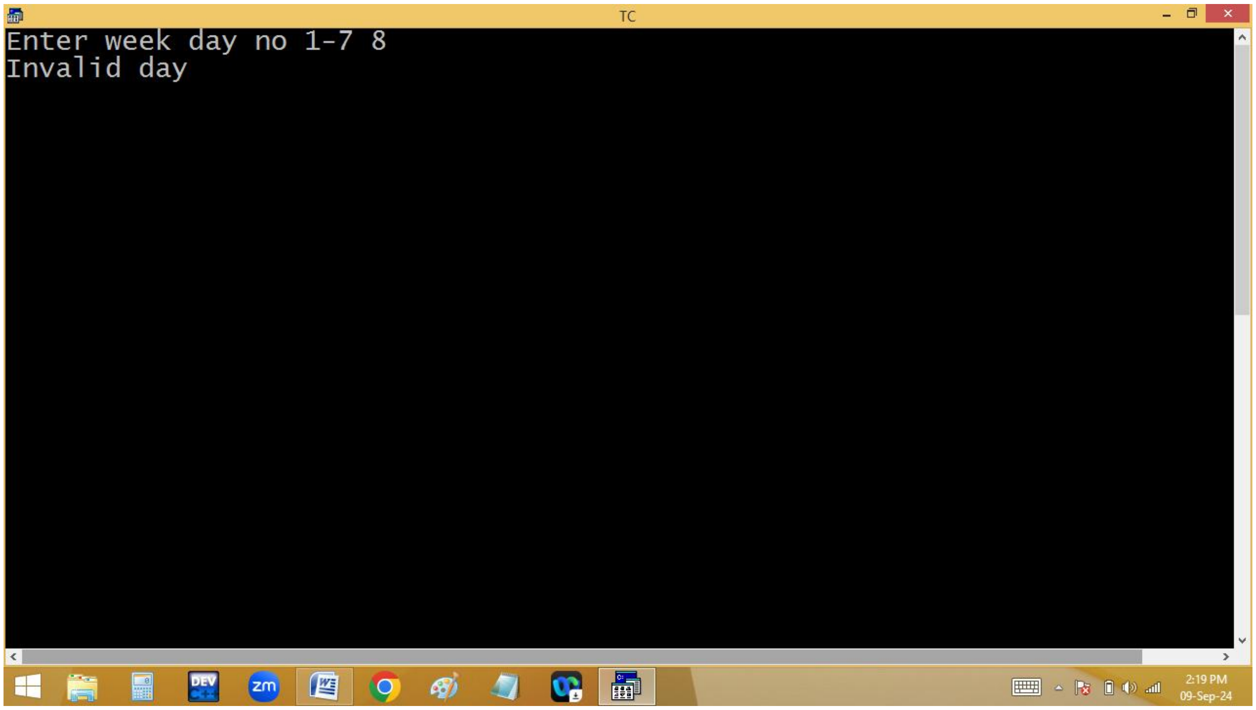


Finding weekday name:


TC
File Edit Run Compile Project Options Debug Break/watch
Line 14 Col 27 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter week day no 1-7 ");scanf("%d",&n);
switch(n)
{
case 1: puts("Sunday");break;
case 2: puts("Monday");break;
case 7: puts("Saturday");break;
default: puts("Invalid day");
}
getch();
}
2:19 PM
09-Sep-24

TC
Enter week day no 1-7 1
Sunday
2:19 PM
09-Sep-24


TC
Enter week day no 1-7 2
Monday
2:19 PM
09-Sep-24


TC
Enter week day no 1-7 7
Saturday
2:19 PM
09-Sep-24



Hotel bill generation:

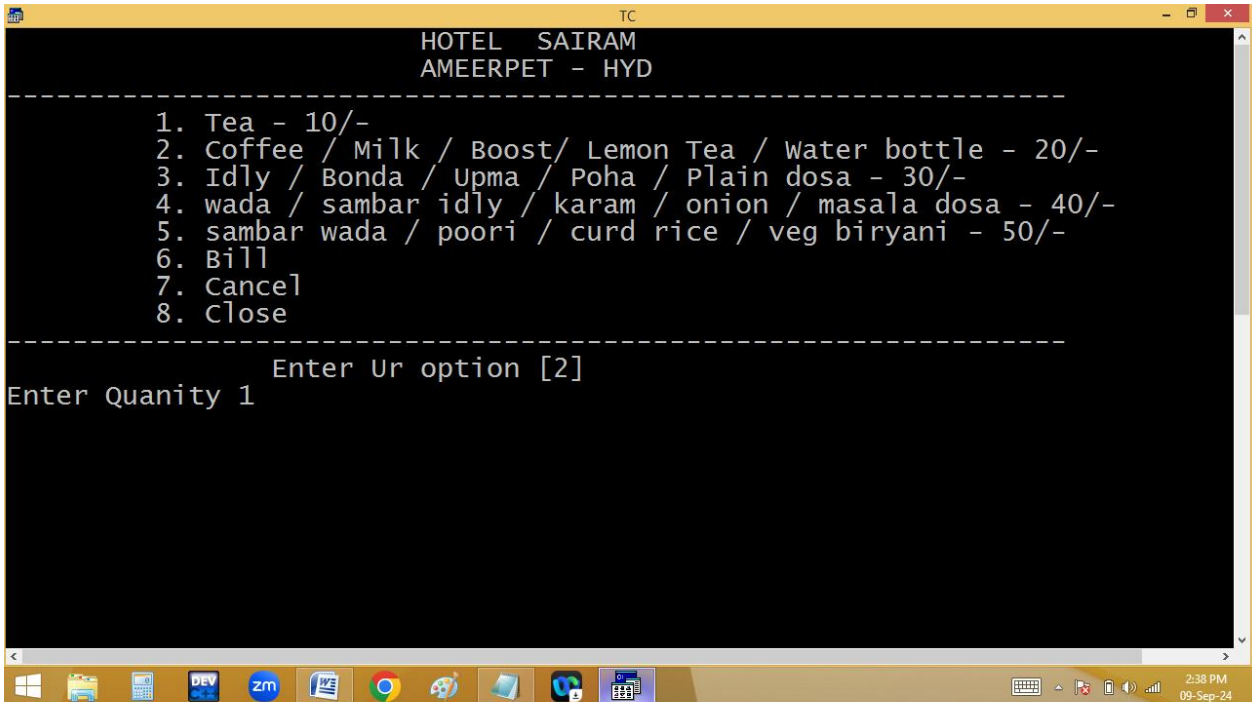
```
#include<stdio.h>

#include<conio.h>

void main()

{

int qty,op;

float amt=0;

start:

clrscr();

puts("\t\t\t HOTEL  SAIRAM");

puts("\t\t\t AMEERPET - HYD");
```

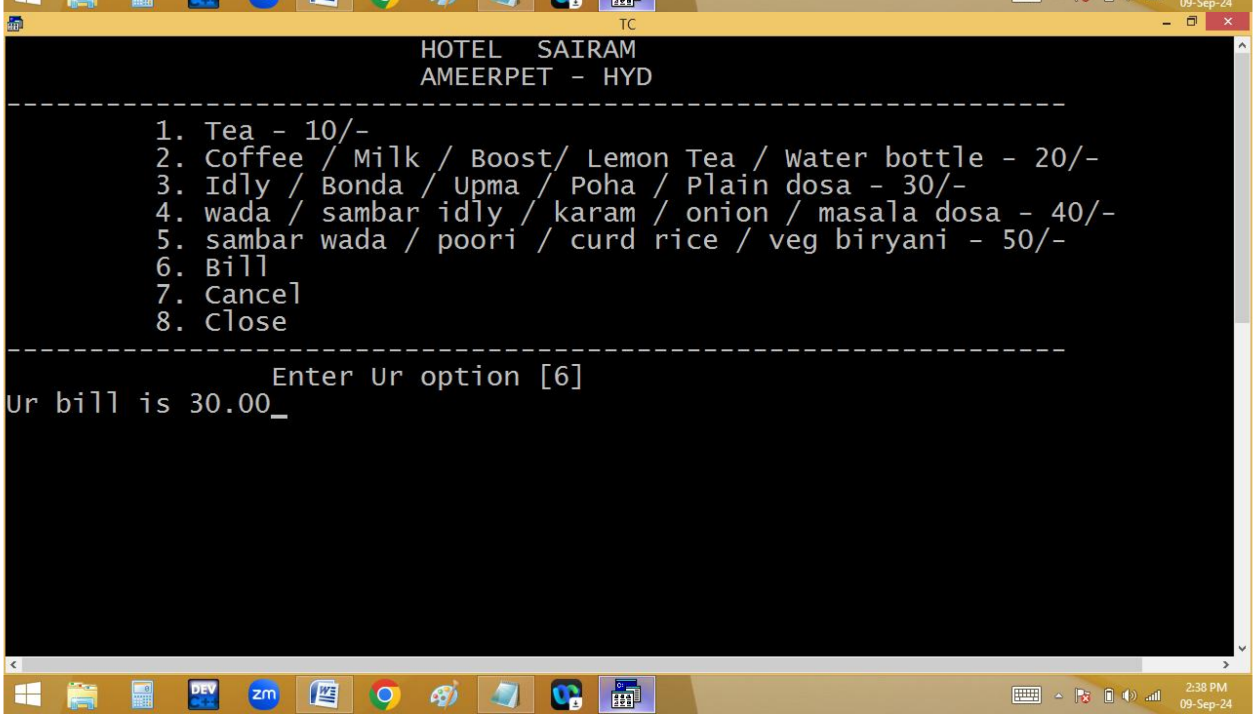
```
puts("---------------------------------------------------------------");
puts("\t 1. Tea - 10/-");
puts("\t 2. Coffee / Milk / Boost/ Lemon Tea / Water bottle - 20/-");
puts("\t 3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-");
puts("\t 4. wada / sambar idly / karam / onion / masala dosa - 40/- ");
puts("\t 5. sambar wada / poori / curd rice / veg biryani - 50/-");
puts("\t 6. Bill");
puts("\t 7. Cancel");
puts("\t 8. Close");
puts("---------------------------------------------------------------");
printf("\t\tEnter Ur option [ ]\b\b") ;
scanf("%d",&op);
if(op<=5) {printf("Enter Quanity "); scanf("%d",&qty);}
switch(op)
{
case  1: amt+=qty*10;break;
case  2: amt+=qty*20;break;
```

```
case  3: amt+=qty*30;break;

case  4: amt+=qty*40;break;

case  5: amt+=qty*50;break;

case  6: printf("Ur bill is %.2f",amt); amt=0;getch(); break;

case  7: printf("Ur order cancelled",amt=0);getch();break;

case  8: return;

}

goto start;

}
```

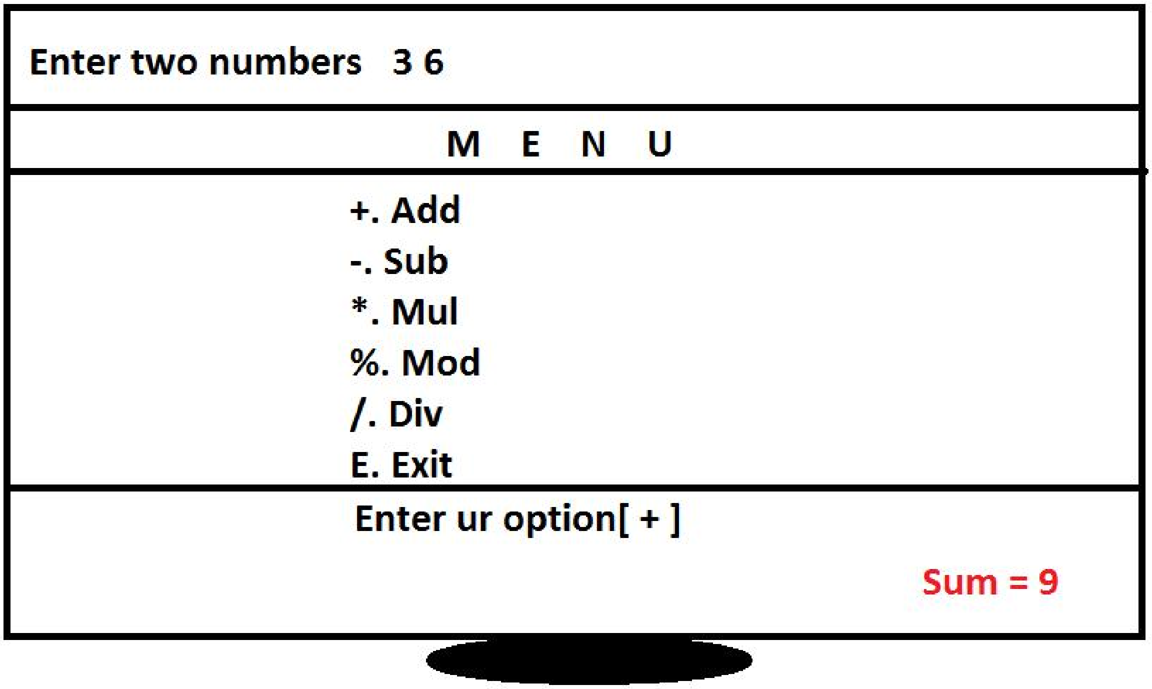
```
                    HOTEL   SAIRAM
                    AMEERPET - HYD
----------------------------------------------------------------
         1. Tea - 10/-
         2. Coffee / Milk / Boost/ Lemon Tea / Water bottle - 20/-
         3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-
         4. wada / sambar idly / karam / onion / masala dosa - 40/-
         5. sambar wada / poori / curd rice / veg biryani - 50/-
         6. Bill
         7. Cancel
         8. Close
----------------------------------------------------------------
                Enter Ur option [8]
```

```
                    HOTEL   SAIRAM
                    AMEERPET - HYD
----------------------------------------------------------------
         1. Tea - 10/-
         2. Coffee / Milk / Boost/ Lemon Tea / Water bottle - 20/-
         3. Idly / Bonda / Upma / Poha / Plain dosa - 30/-
         4. wada / sambar idly / karam / onion / masala dosa - 40/-
         5. sambar wada / poori / curd rice / veg biryani - 50/-
         6. Bill
         7. Cancel
         8. Close
----------------------------------------------------------------
                Enter Ur option [1]
Enter Quanity 1_
```





Read two numbers and perform arithmetic operation using menu:



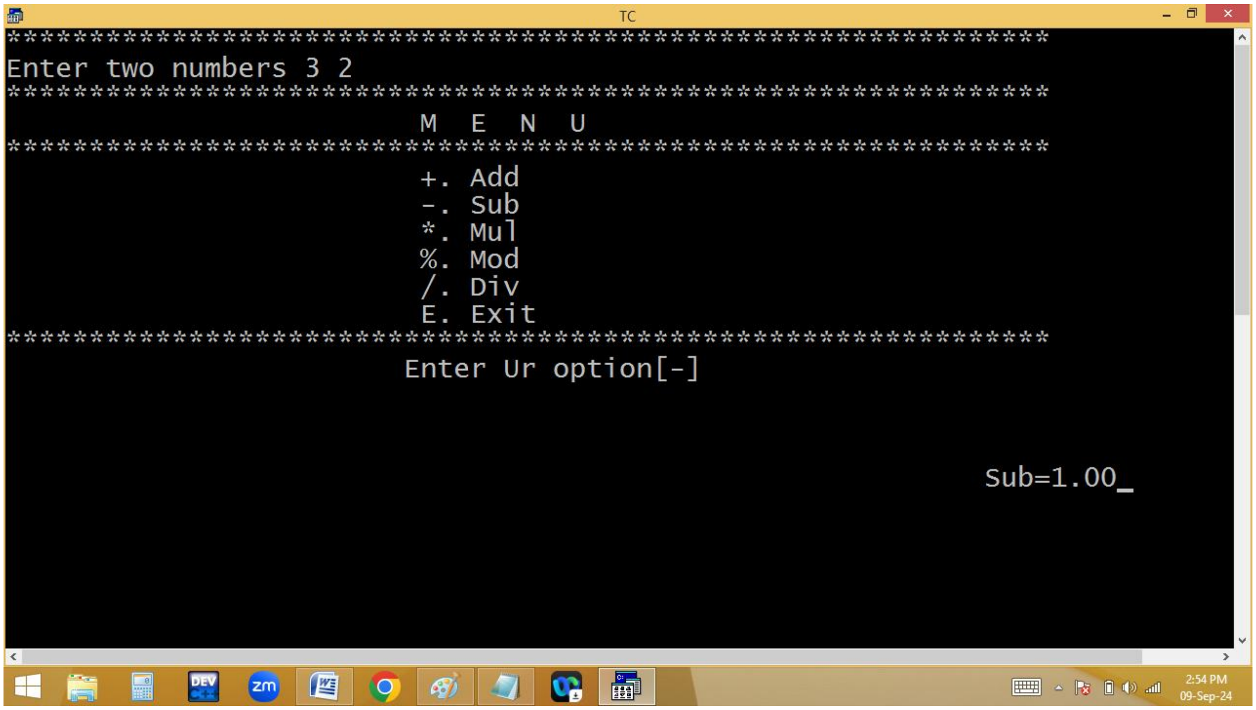

```
#include<stdio.h>

#include<conio.h>

#include<math.h>

void main()

{

float a,b;

char op;

start:

clrscr();

puts("***********************************************
****************");
```

```
printf("Enter two numbers "); scanf("%f %f",&a, &b);

puts("***********************************************
****************");

puts("\t\t\t M  E  N  U");

puts("***********************************************
****************");

puts("\t\t\t +. Add");

puts("\t\t\t -. Sub");

puts("\t\t\t *. Mul");

puts("\t\t\t %. Mod");

puts("\t\t\t /. Div");

puts("\t\t\t E. Exit");

puts("***********************************************
****************");

printf("\t\t\tEnter Ur option[ ]\b\b");scanf(" %c",&op);

gotoxy(60,17); /* 60-col, 17-row */

switch(op)

{

case '+': printf("Sum=%.2f",a+b);break;
```

```
case '-': printf("Sub=%.2f",a-b);break;

case '*': printf("Mul=%.2f",a*b);break;

case '%': printf("Mod=%.2f",fmod(a,b));break;

case '/': printf("Div=%.2f",a/b);break;

case 'e': case 'E': return;

}

getch();

goto start;

}
```

```
TC
***************************************************************
Enter two numbers 2 3
***************************************************************
                         M  E  N  U
***************************************************************
                         +. Add
                         -. Sub
                         *. Mul
                         %. Mod
                         /. Div
                         E. Exit
***************************************************************
                        Enter Ur option[E]
```

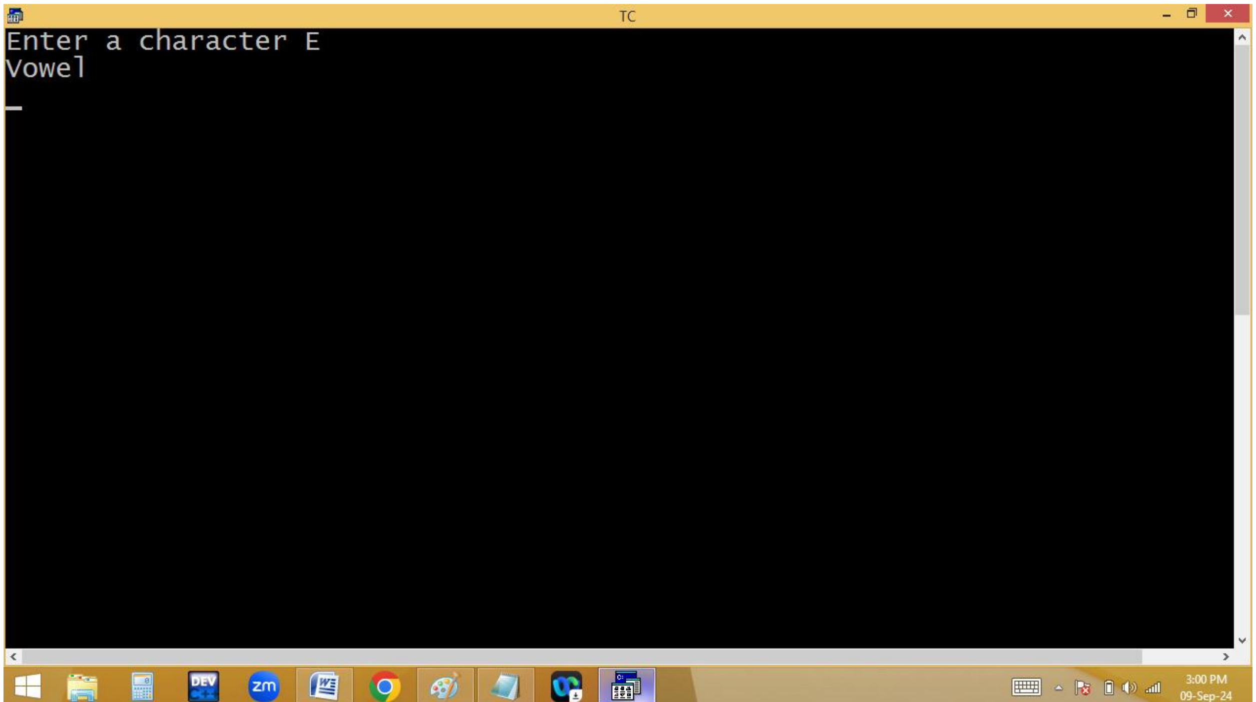
```
TC
***************************************************************
Enter two numbers 3.3 5.4
***************************************************************
                         M  E  N  U
***************************************************************
                         +. Add
                         -. Sub
                         *. Mul
                         %. Mod
                         /. Div
                         E. Exit
***************************************************************
                        Enter Ur option[*]

                                                           Mul=17.82_
```

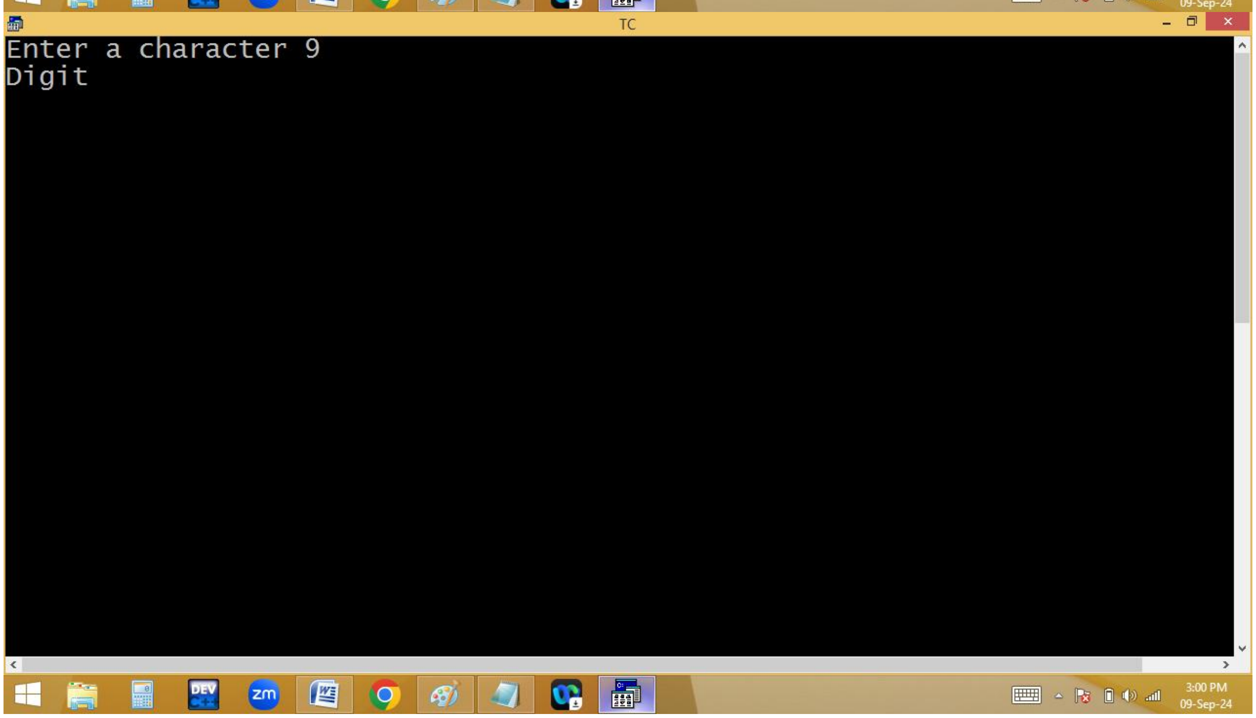


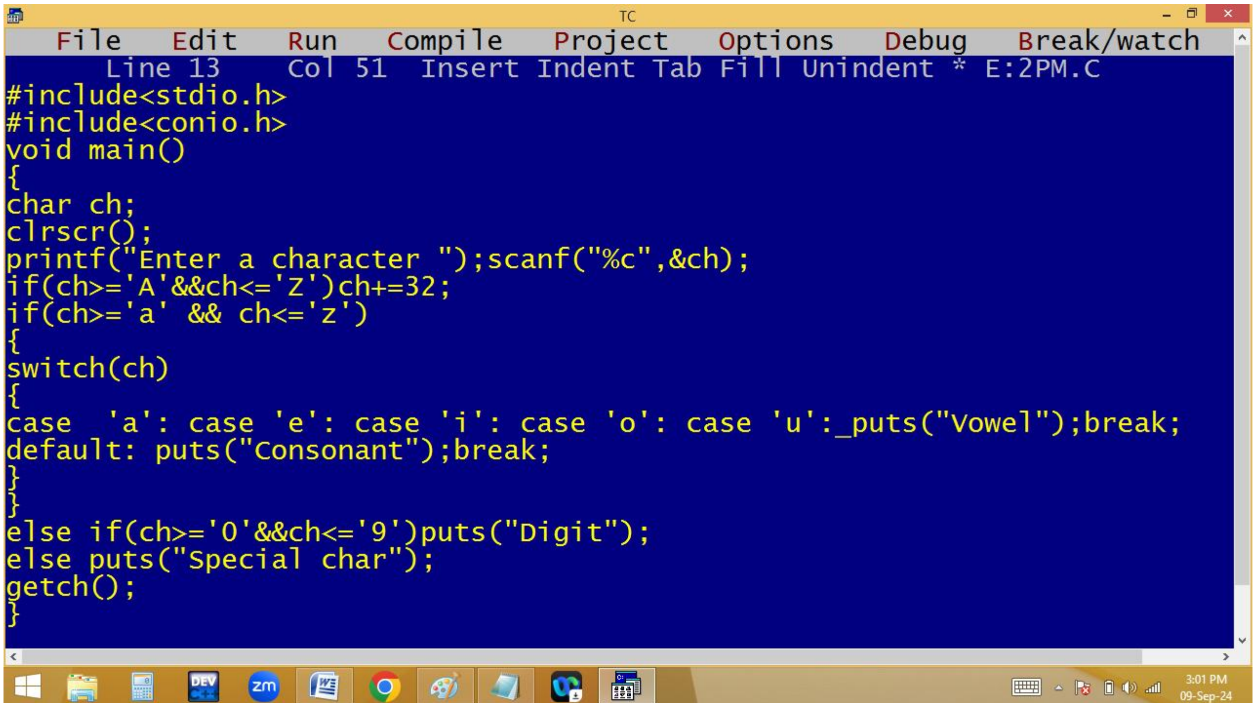
Finding vowel/consonant using switch:

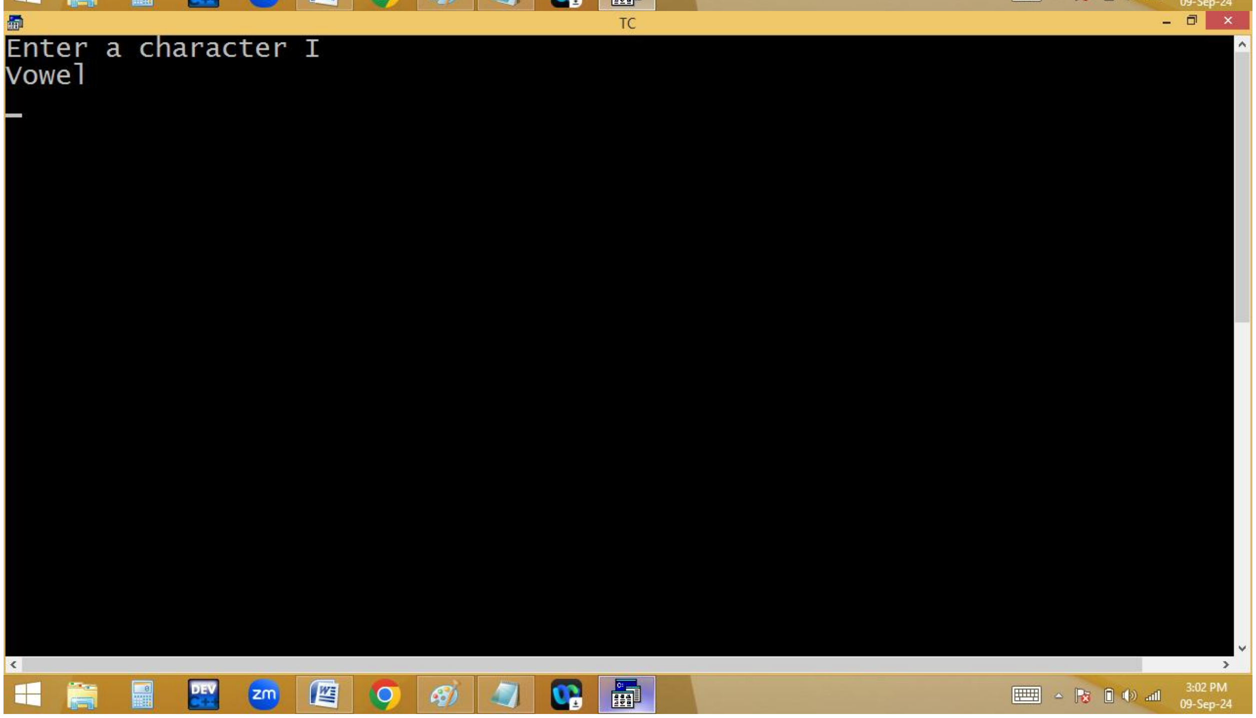
```
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 19    Col 27  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='A'&&ch<='Z')ch+=32;
if(ch>='a' && ch<='z')
{
switch(ch=='a'||ch=='e'||ch=='i'||ch=='o'||ch=='u')
{
case  1: puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(ch>='0'&&ch<='9')puts("Digit");
else puts("Special char");
getch();
}
```

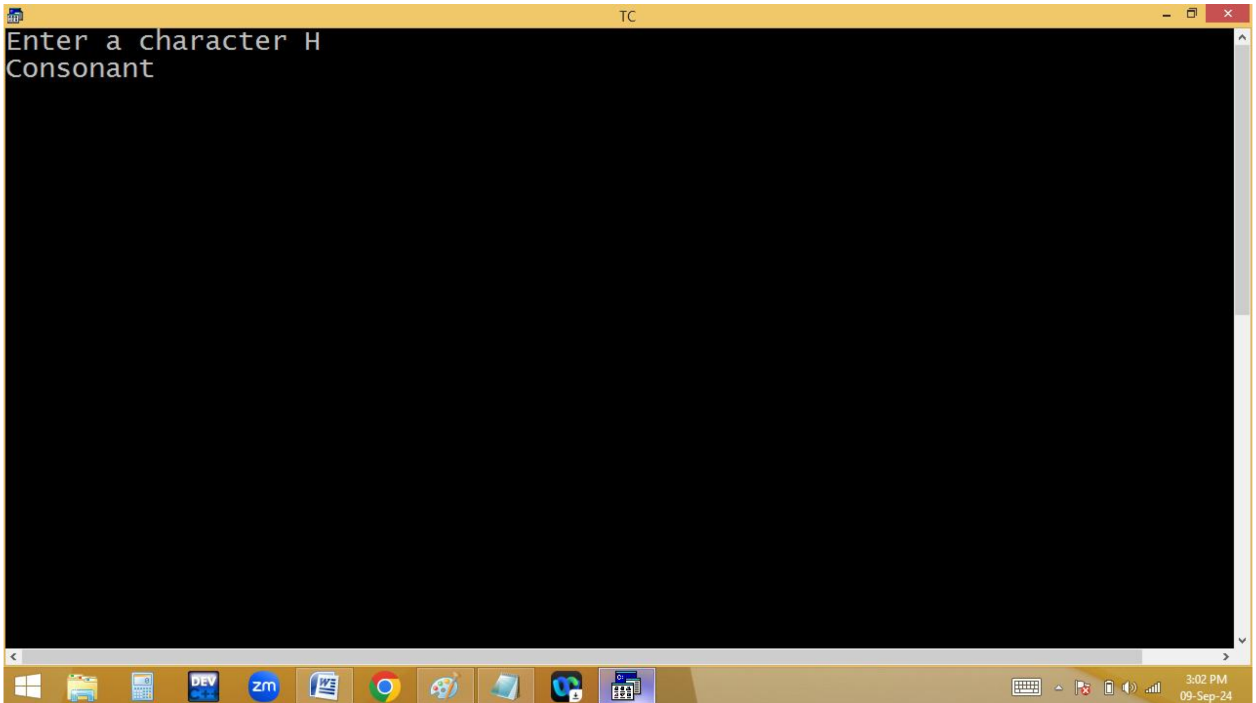
```
Enter a character h
Consonant
```

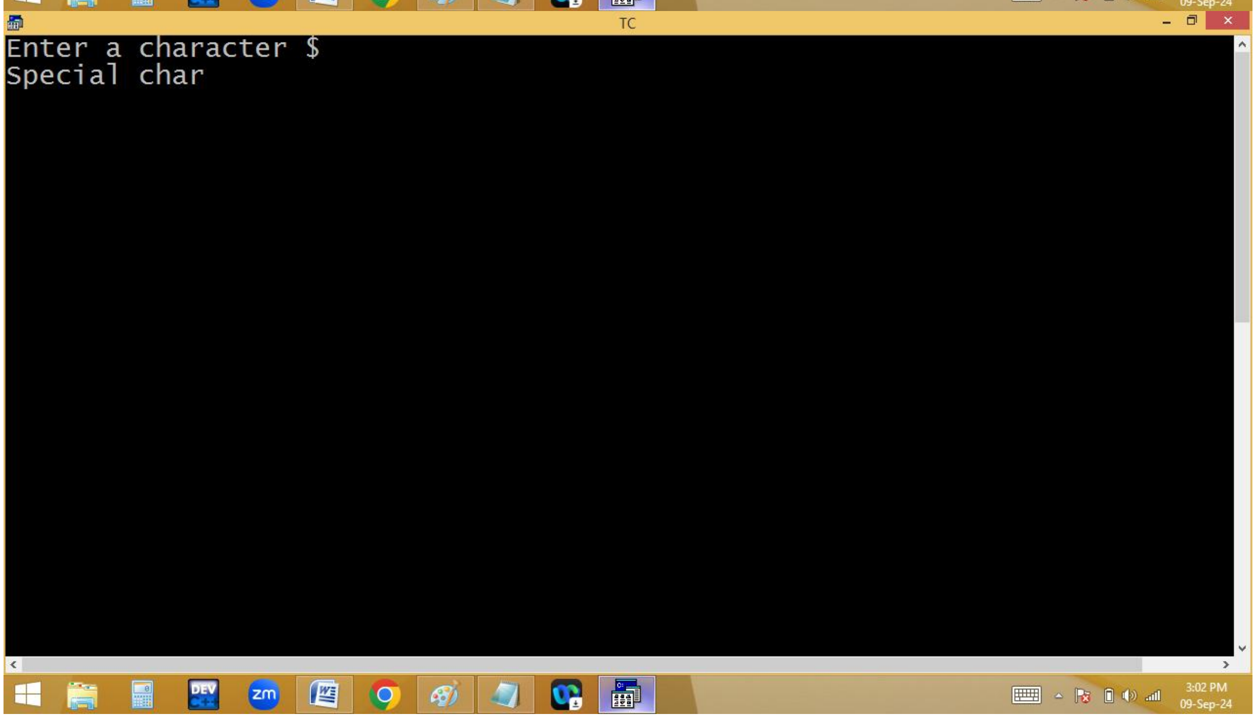

TC
Enter a character E
Vowel
3:00 PM
09-Sep-24


TC
Enter a character 9
Digit
3:00 PM
09-Sep-24


TC
File Edit Run Compile Project Options Debug Break/watch
Line 13 Col 51 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(ch>='A'&&ch<='Z')ch+=32;
if(ch>='a' && ch<='z')
{
switch(ch)
{
case 'a': case 'e': case 'i': case 'o': case 'u': puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(ch>='0'&&ch<='9')puts("Digit");
else puts("Special char");
getch();
}
3:01 PM
09-Sep-24


TC
Enter a character I
Vowel
3:02 PM
09-Sep-24

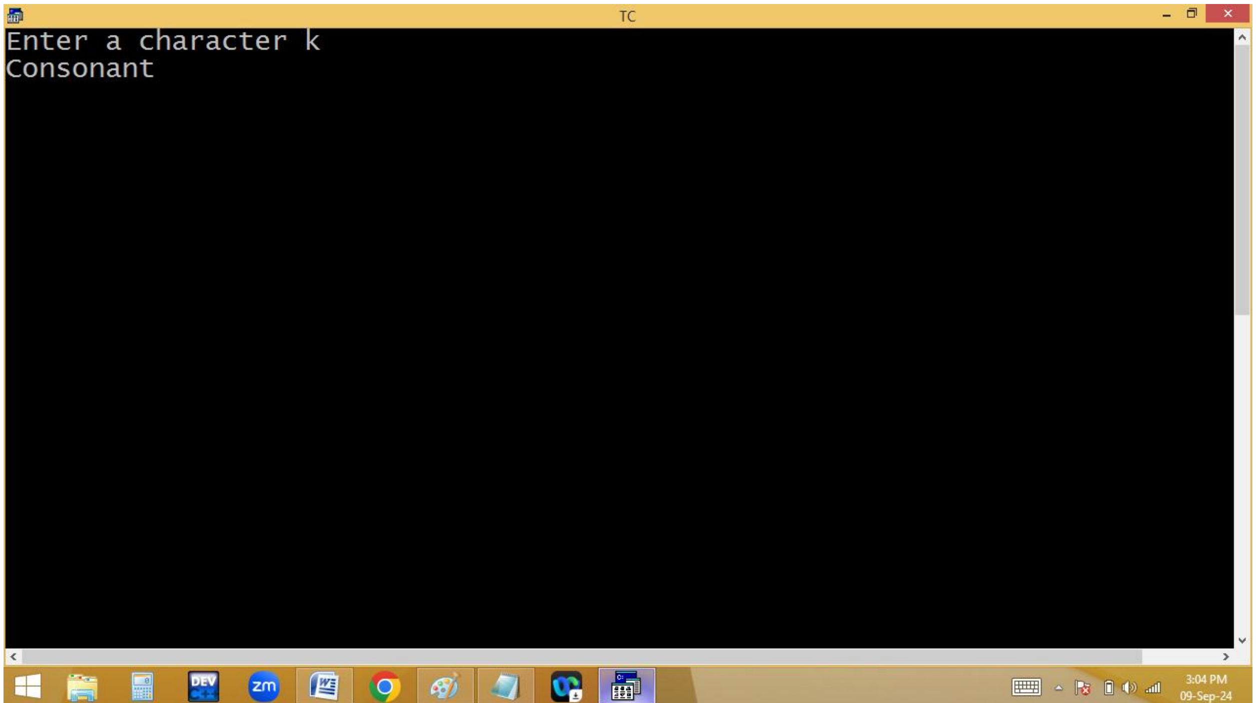
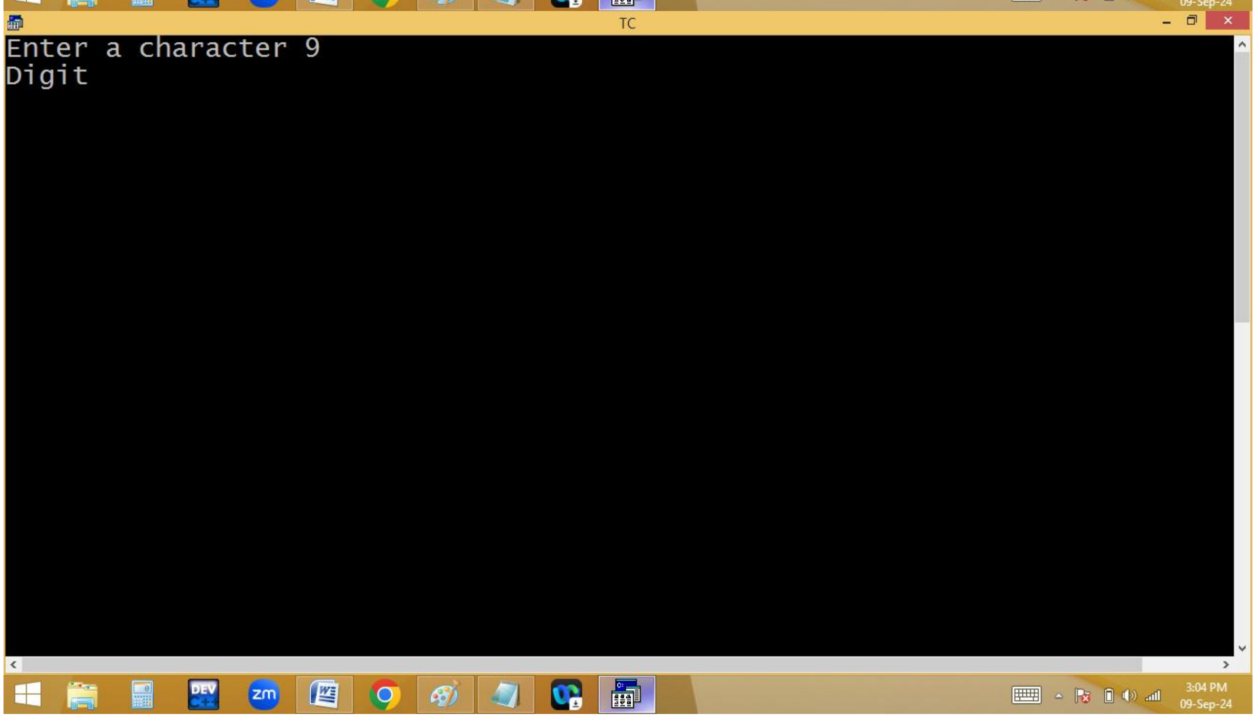


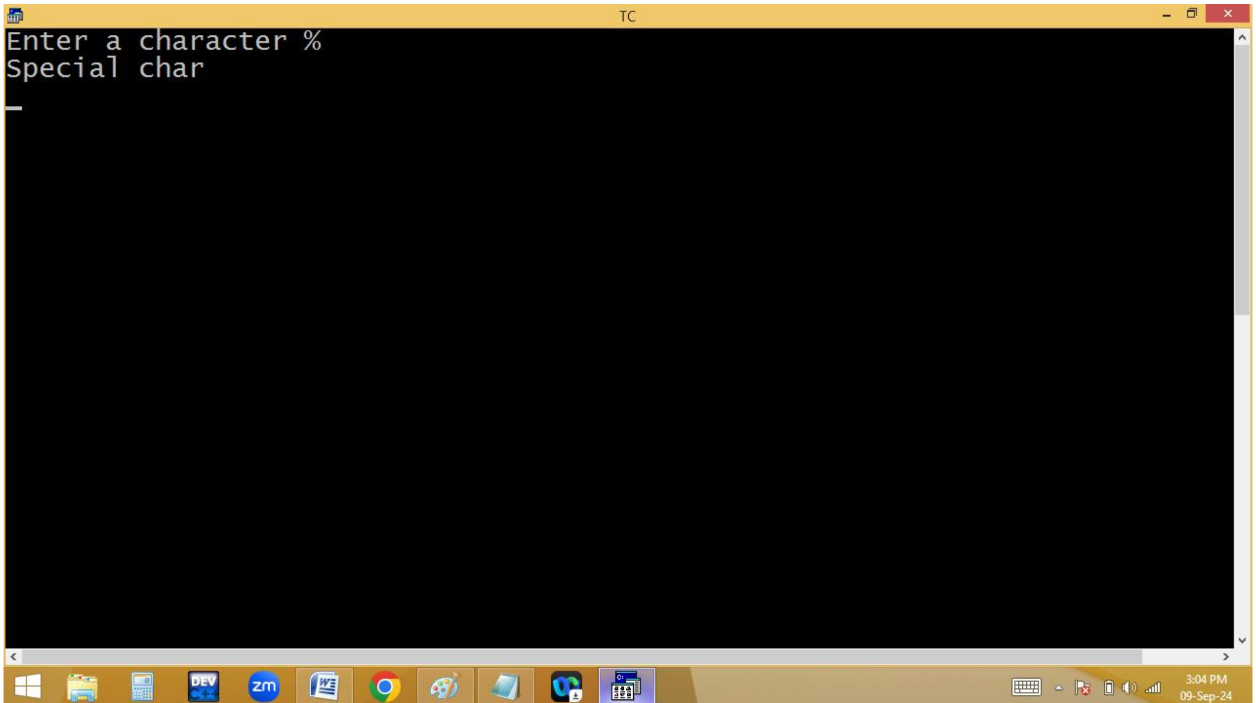


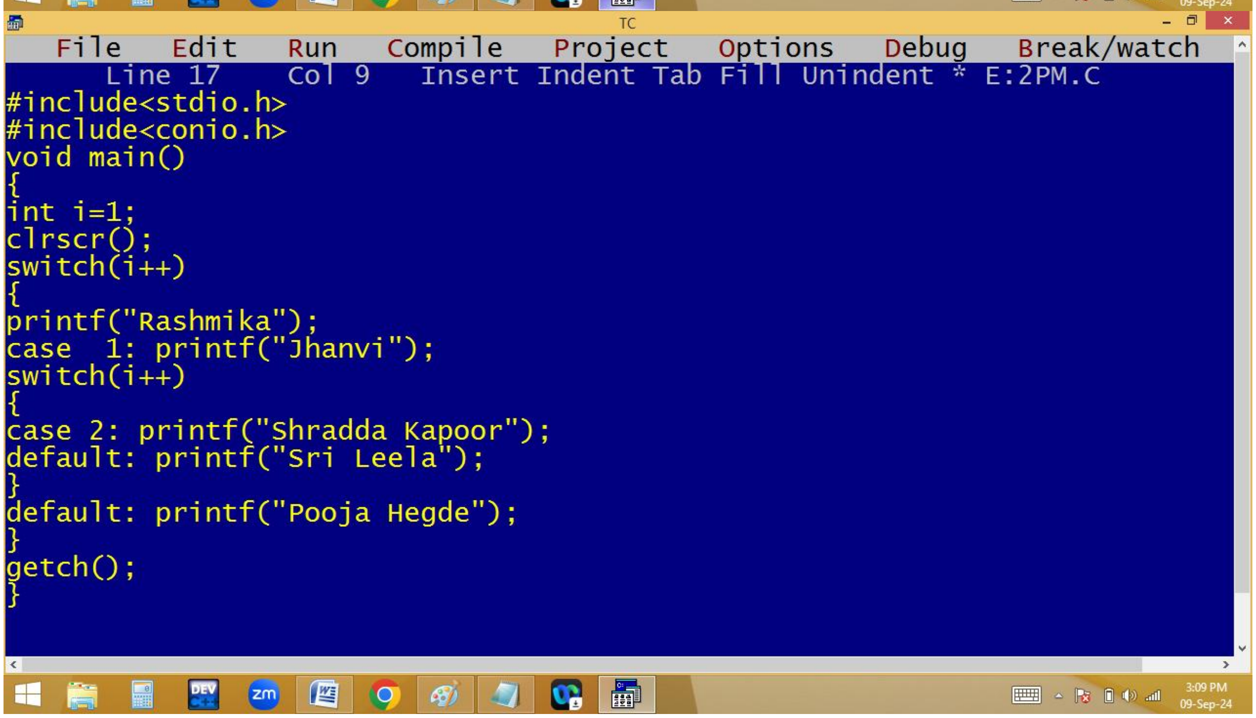
Using predefined functions:

```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter a character ");scanf("%c",&ch);
if(isupper(ch))ch=tolower(ch);
if(isalpha(ch))
{
switch(ch)
{
case  'a': case 'e': case 'i': case 'o': case 'u': puts("Vowel");break;
default: puts("Consonant");break;
}
}
else if(isdigit(ch))puts("Digit");
else puts("Special char");
getch();
}
```

```
Enter a character o
Vowel
```


TC
Enter a character k
Consonant
3:04 PM
09-Sep-24

TC
Enter a character 9
Digit
3:04 PM
09-Sep-24


TC
Enter a character %
Special char
3:04 PM
09-Sep-24

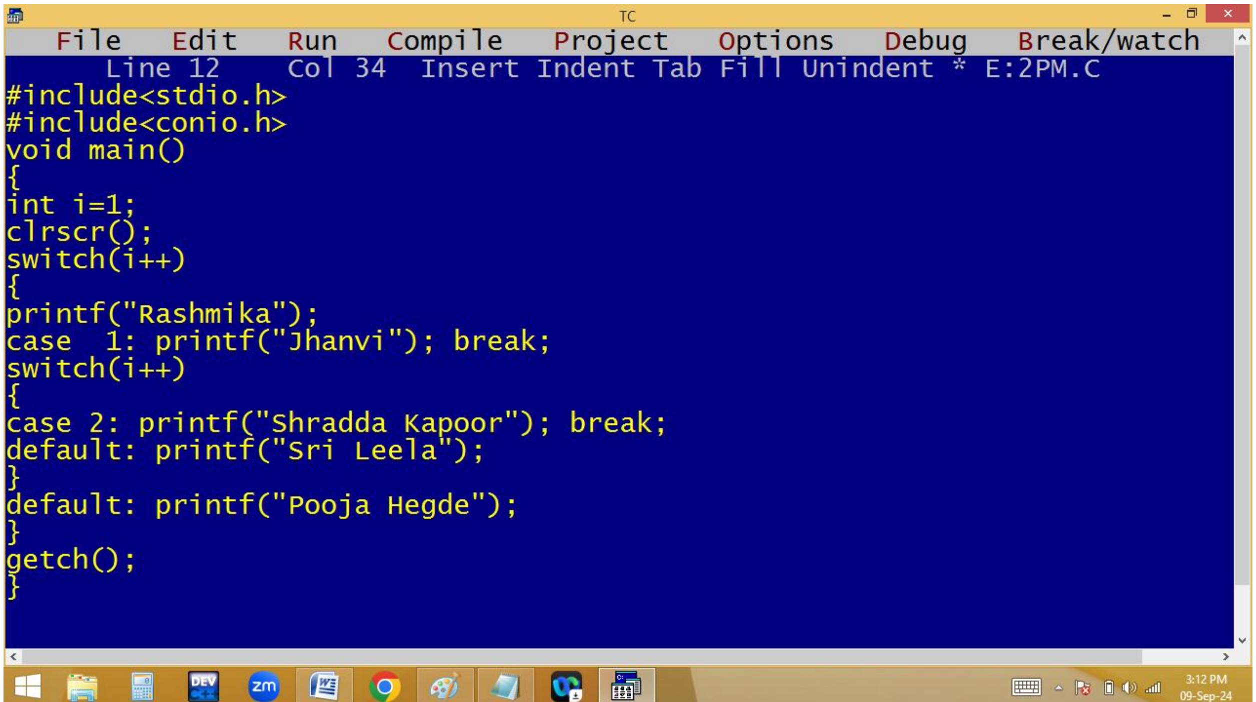

TC
File Edit Run Compile Project Options Debug Break/watch
Line 17 Col 9 Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
switch(i++)
{
printf("Rashmika");
case 1: printf("Jhanvi");
switch(i++)
{
case 2: printf("Shradda Kapoor");
default: printf("Sri Leela");
}
default: printf("Pooja Hegde");
}
getch();
}
3:09 PM
09-Sep-24

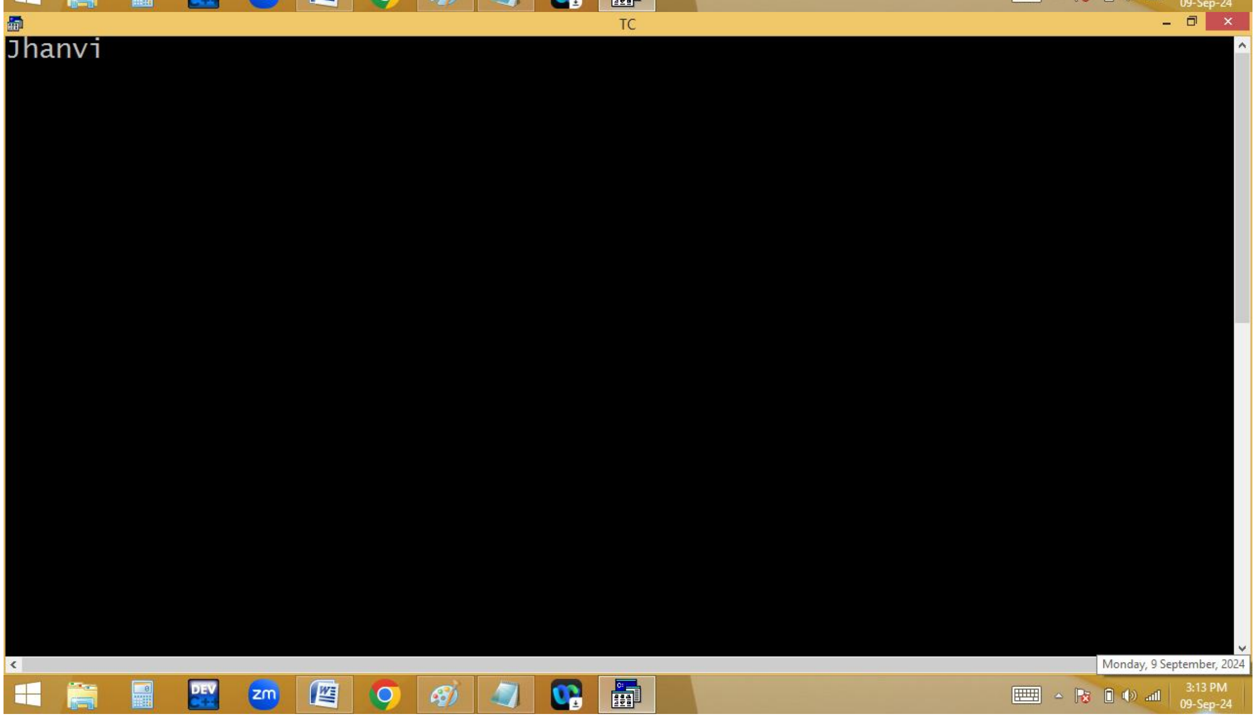
```
JhanviShradda KapoorSri LeelaPooja Hegde
```

```
File  Edit  Run  Compile  Project  Options  Debug  Break/watch
Error: Misplaced continue in function main
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
switch(i++)
{
printf("Rashmika");
case  1: printf("Jhanvi");  continue;
switch(i++)
{
case 2: printf("Shradda Kapoor");
default: printf("Sri Leela");
}
default: printf("Pooja Hegde");
}
getch();
}
/* error */
```

```
File    Edit    Run    Compile    Project    Options    Debug    Break/watch
      Line 20     Col 1    Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int i=1;
clrscr();
switch(i++)
{
printf("Rashmika");
case  1: printf("Jhanvi");
switch(i++)
{
case 2: printf("Shradda Kapoor"); break;
default: printf("Sri Leela");
}
default: printf("Pooja Hegde");
}
getch();
}
```

```
JhanviShradda KapoorPooja Hegde_
```





Loops / Iterations / Repetitive statements

Loops are used to repeat a block/group of statements continuously until the given condition becomes false.

Loops reduce program size and improves performance.

In loops beginning and ending points are same.

C-Language supports basically 2 types of loops.

1. **Entry/pre controlled loops.**
2. **Exit/post controlled loops.**

In entry control loops, condition is tested first and it is true then only statements block is executed.

Under entry control loops we are having

i. While loop

ii. For loop

In exit control loop, the statements are executed first and later condition is tested.

Under exit control loop we are having

i. do while.

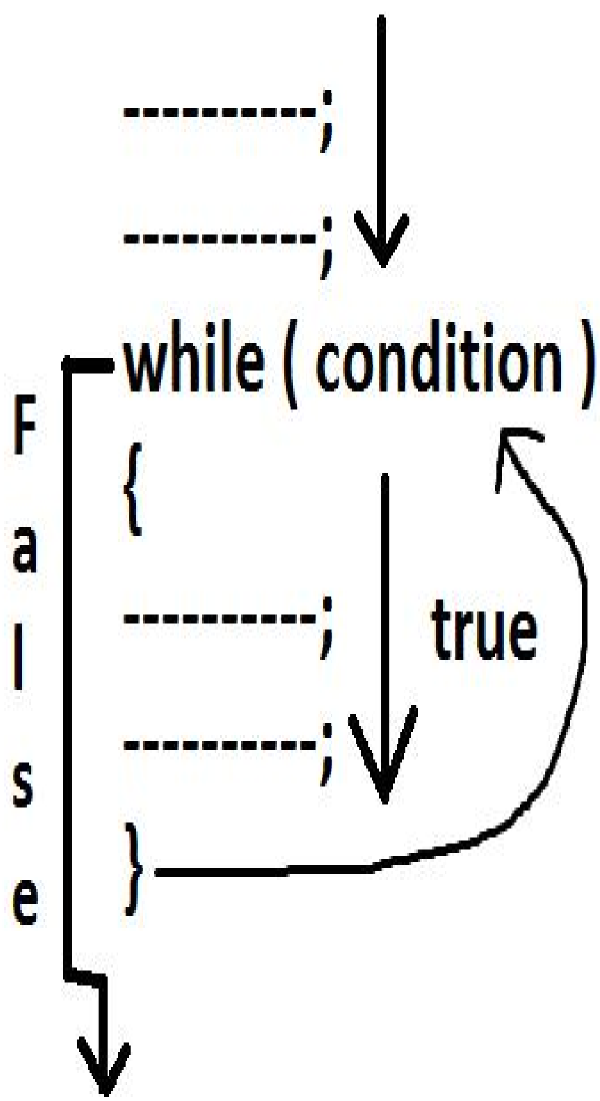
While loop:

- while is a keyword.
- In while loop condition is tested first and it is true then only while block statements are executed. After executing while block statements, the program execution automatically shifted/jumped to while condition at the beginning. If it is true then once again the while block statements are repeated. Like this the

process is continued until while condition becomes false.

- **While is entry control loop.**

Syntax:



Flow chart: